

OLIST BRAZILIAN E-COMMERCE SALES ANALYSIS REPORT

Project Overview

This project analyzes the Olist Brazilian E-Commerce dataset to understand sales performance, customer behavior, product trends, and regional performance. The data was cleaned, transformed, and visualized using Python libraries such as Pandas, Matplotlib, and Seaborn.

Objectives

- Identify top-performing product categories.
- Analyze monthly sales trends.
- Evaluate regional sales performance.
- Understand customer purchasing patterns.
- Examine customer review behavior.
- Generate actionable business insights.

Data Preparation

The dataset was cleaned by:

- Handling missing values in product categories.
- Removing duplicate review records.
- Converting purchase timestamps into datetime format.
- Merging customers, orders, products, reviews, and order items datasets into a single analytical dataset.
- Creating a revenue column using product prices.

Visualizations Created

1. Revenue by Product Category (Bar Chart)
2. Monthly Sales Trend (Line Chart)
3. Revenue by State (Bar Chart)
4. Order Value Distribution (Histogram)
5. Review Score Distribution (Pie Chart)
6. Category vs Month Revenue Heatmap

Key KPIs

- Total Revenue
- Total Orders
- Top Performing Product Category
- Best Sales Month

Conclusion

The analysis highlights the categories, regions, and periods that generate maximum revenue. Customer reviews indicate overall satisfaction levels, while sales trends provide insights for inventory planning and marketing campaigns. The findings can help management improve profitability and customer experience.